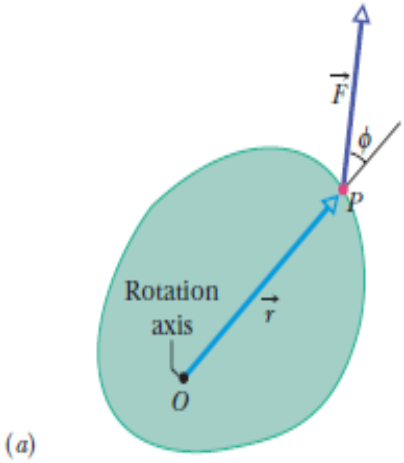
Torque

- Torque: is defined as the tendency of a force to produce rotation about an axis. The rotational equivalent of force. It is the effectiveness of a force at causing rotation. It is given by the product of the perpendicular force and the distance (r) from the point of application to the axis of rotation



$$\tau_a = \vec{F} \times \vec{r} \quad , \quad \tau_a = Fr \sin\phi$$

- r: called the lever arm or moment arm (the distance from the pivot point to the point , distance from axis) it is the perpendicular distance between the rotation axis and an extended line running through the vector), F: applied force on the lever arm,ϕ: angle between force vector and lever arm.

- F: is the magnitude of the force vector. Only the component of the force perpendicular to the lever arm ($F \sin\phi$) causes rotation. A force applied directly toward or away from the pivot (ϕ= 0) produces zero torque
- The SI unit of torque is the newton-meter (Nm).
- The larger the lever arm or force, the greater the torque.
- A torque t is positive if it tends to rotate a body at rest counterclockwise and negative if it tends to rotate the body clockwise.
- Newton's Second Law for Rotation

$$F_{net} = \mathrm{ma}$$

& $$\overrightarrow{F_{net}} = \frac{\mathrm{d}\vec{p}}{dt} = \mathrm{m}\frac{\mathrm{d}\vec{v}}{dt} = m\vec{a}$$

The net torque on a rigid object causes it to undergo an angular acceleration.

The net external force on the object must equal zero.

$$\sum \vec{F}_{ext} = 0$$

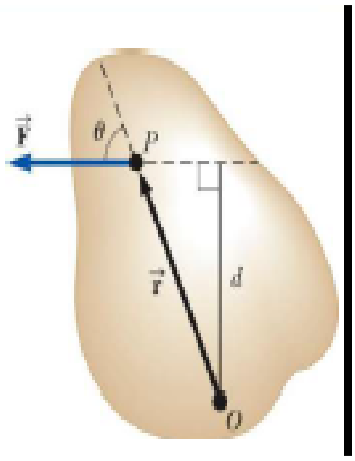
If the object is modeled as a particle, then this is the only condition that must be satisfied.

The net external torque on the object about any axis must be zero.

$$\sum \vec{\tau}_{ext} =\cdot 0$$

This is needed if the object cannot be modeled as a particle.

These conditions describe the rigid object in equilibrium analysismodel.



We will restrict the applications to situations in which all the forces lie in the x-y plane. There are three resulting equations:

$$\sum F_x = 0, \sum F_y = 0, \sum F_z = 0,$$

- Moment of Inertia (I): It is the rotational equivalent of mass. While mass measures resistance to linear acceleration, the moment of inertia measures resistance to angular acceleration.

$$I = \sum m_i r_i^2$$

- The sum of the mass of each particle times the square of its distance from the axis of rotation.
- The distribution of mass relative to the pivot point is critical. Mass farther from the axis greatly increases I, making it harder to start or stop rotation.

- Apply Newton's second law for rotation to relate the net torque on a body to the body's rotational inertia and rotational acceleration, all calculated relative to a specified rotation axis. :

$$\Sigma\tau_a = I\alpha$$

- The net torque on an object is equal to its moment of inertia times its angular acceleration.
- τ_a: The net torque acting on a particle or rigid body, (I) is the rotational inertia of the particle or body about the rotation axis, and (α) is the resulting angular acceleration about that axis.
- This is the direct analog of $\sum F = ma$

Work and Power in Rotational Motion:

All work and energy concepts have rotational analogs.

- Work done by a Torque (W):

$$W = \int \tau \, \mathrm{d}\theta$$

For constant torque: $\mathrm{W} = \tau\Delta\theta$

  - Analog of $\mathrm{W} = \int F dx$
  - Work-Kinetic Energy Theorem for Rotation:
  - $\mathrm{W} = \Delta k = -(\frac{1}{2} I\, \omega_f{}^2 - \frac{1}{2} I\, \omega_i{}^2)$

- Power (P):

$$p = \frac{dW}{dt} = \tau\omega$$

Analog of $P = Fv$

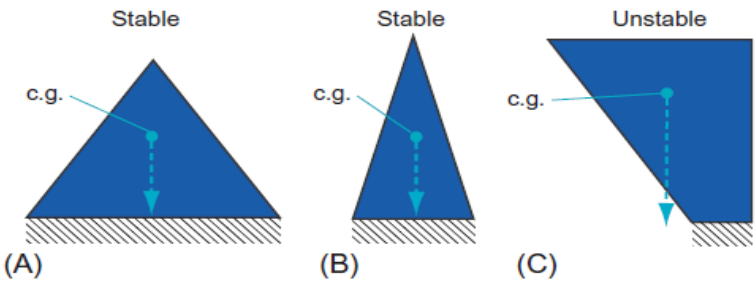
- Biological Context: The Mechanics of Forearm Lifting (Biophysics)
- The Driving Force (from the Muscle): To lift your forearm, your bicep muscle contracts. This muscle pulls on the forearm bone at a point very close to the elbow joint, which acts as the pivot. Because the lever arm (the distance from the pivot to the point of force application) here is short, the muscle must exert a very large force to generate sufficient torque to lift the forearm.
- The Opposing Force (Weight of the Forearm): The weight of the forearm itself acts as an opposing force. The center of mass of the forearm is located farther away from the elbow pivot. This means it has a longer lever arm. Even though the weight of the forearm itself might be small, this long lever arm creates a significant opposing torque.

- Conclusion: For the driving force (the muscle) to overcome the opposing force (the weight of the forearm), your bicep muscle must exert a force much larger than the weight of your arm itself to compensate for its short lever arm.

Equilibrium and Stability

1. Basic Concept

   - Center of Gravity (COG): is the point where the total gravitational force (weight) on the body is considered to act. If you suspend an object from this point, it will balance and not rotate.
   - Center of Mass (COM): is the point that represents the average position of the mass of the body. The average location of all the mass in a body. It is the point that moves as if all the mass of the body were concentrated there. It's the point where we can consider the entire mass to be concentrated for analyzing its motion (e.g., the path of a projectile).
   - Base of Support: Area of contact between the body and supporting surface
   - Stable Equilibrium: When the center of gravity is above the base of support



1.1 Stability of bodies.

2. The Relationship Between COM and COG:

Under Normal Conditions on Earth: If the acceleration due to gravity (g) is the same for all parts of a body, the center of gravity and center of mass for the same object are identical.
Why? Because the acceleration due to gravity (g) is essentially constant for all parts of the object. When (g) is uniform, the location where the total gravitational force effectively acts (the COG) coincides exactly with the average location of the mass (the COM).

Center of Gravity (COG) and Center of Mass (COM) (Conts.)

- When Do They Differ?
  The center of gravity differs from the center of mass only when the object is so large that the acceleration due to gravity (g) varies significantly between its different parts, such as an object extending from the Earth's surface into space. For everyday objects, they are considered the same point.
- Example: A gigantic object extending from the Earth's surface into space. The gravitational force on parts closer to Earth is stronger than on parts farther away. In this rare case, the center of gravity would be slightly closer to Earth than the center of mass.
- Conclusion: For all practical purposes and everyday objects (like a car, a chair, or the human body), we treat the center of mass and the center of gravity as the same point.
- 3. Equilibrium Conditions:
- An object in static equilibrium if both its linear (net force) or at rest and angular acceleration (net torque) or not rotating are zero.
- Conditions for Static Equilibrium:
  1. Translational Equilibrium: The net force on the object must be zero. This ensures there is no linear acceleration.
  2. Rotational Equilibrium: The net torque on the object (about any chosen pivot point) must be zero. This ensures there is no angular acceleration.

To satisfy this condition, the sum of the forces F acting on the body, as well as the sum of the torques τ_a produced by these forces must be zero; that is,

$$\sum F = 0 \quad and \quad \sum \tau_a = 0$$

- 4. Factors Affecting Stability:
  - Base width: Wider base increases stability The wider the base on which the body rests, the more stable it is; that is, the more difficult it is to topple it. If the wide-based body. (Fig. A)
  - Center of gravity height: Lower center of gravity increases stability
- Biological Context: This is the basis for understanding posture and the forces acting on bones and joints. For example, it allows calculation of the massive force the hip abductor muscles must exert to keep the pelvis level when you stand on one leg.
- In human biology:

- The body maintains equilibrium through muscle control and sensory feedback (especially from the inner ear).
- Physical therapists use torque analysis to understand posture and balance.
- When standing upright, the line of gravity must pass through the center of support — otherwise, you tip over.

Center of Gravity (COG) and Center of Mass (COM) (Conts.)

- Proof:
  - A body is composed of many small elements, each with mass (m_i)
  - A gravitational force acts on each element: $\overrightarrow{F_{g1}} = m_i \overrightarrow{g_1}$

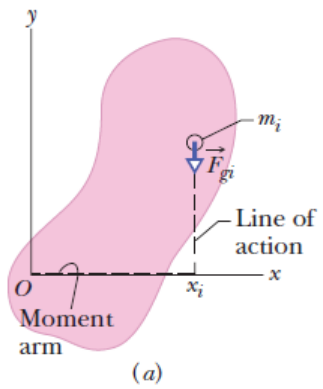


Figure (a): Shows the gravitational force $(\overrightarrow{F_{g1}})$ acting on an element of mass (m_i) within the body. Each force has a moment arm (x_i)

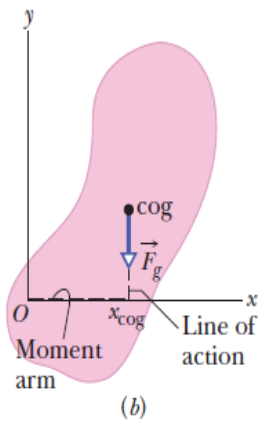


Figure (b): Shows the total gravitational force (weight) $(\overrightarrow{F_{g1}})$ acting at a single point, which is

the Center of Gravity (COG) and coincides with the Center of Mass (COM). This force has a moment arm (x_{cg})

relative to the origin point (O).relative to the origin point (O).

- The net torque of all these forces about an origin (O) is:

$$\tau_{net} = \Sigma x_{\mathrm{i}} F_{gi} = \Sigma x_{\mathrm{i}} m_i g_i$$

- If we assume the total weight force $(\overrightarrow{F_{\mathrm{g}}})$ acts at the center of gravity (x_{cg}), the torque it produces about point (O) is:

$$\tau = x_{\mathrm{cg}} F_{\mathrm{g}}$$

And, since $F_{\mathrm{g}} = \Sigma F_{gi}$then:

$$\tau = x_{\mathrm{cg}} \Sigma F_{gi} = x_{\mathrm{cg}} \Sigma m_i g_i$$

- By the definition of the center of gravity, these two torques are equal:

$$x_{\mathrm{cg}} \Sigma m_i g_i = \Sigma x_{\mathrm{i}} m_i g_i$$

- The Key Idea: If (g_{i}) is the same for all elements (constant g), we can cancel g

from the equation: $x_{\mathrm{cg}} \Sigma m_i = \Sigma x_i m_i$

- And since $\Sigma m_i = M$ (total mass), the equation becomes:

$$x_{\mathrm{cg}} = \frac{1}{M} \Sigma x_i m_i$$

- The right-hand side of this equation is exactly the definition of the center of

mass coordinate x_{com}.Then: $x_{\mathrm{cg}} = x_{\mathrm{com}}$

Static Forces

- **1. Basic Concepts:**
- Biomechanics is the application of mechanical principles to living organisms.
- Static forces: study forces acting on bodies at rest (i.e., when the sum of forces and torques equals zero).
- **Center of Gravity (C.G.):**
  - The point at which the entire weight of a body can be considered to act.

- Affected by the body's mass and its distribution.
- In an erect person, the C.G. is at approximately 56% of total height measured

from the soles of the feet (Fig. 1.3).

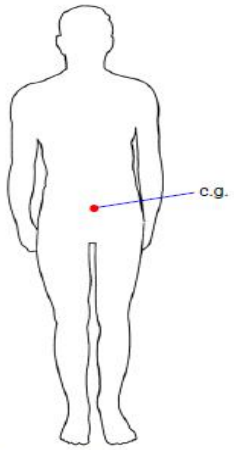


FIGURE 1.3 Center of gravity for a person.

- **Base of Support:**
  - The area bounded by the body's contact points with the ground (e.g., a person's feet).
  - Condition for Stable Equilibrium: The C.G. must be over the base of support (Fig.a, b).
- Stability:
  - Wider base = Greater stability (harder to topple) (Figs. 1.1, 1.2).
  - Lower C.G. = Greater stability.

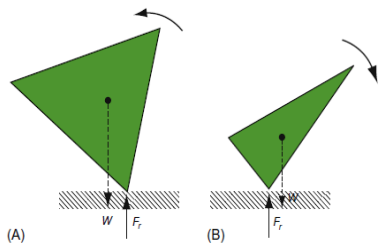


FIGURE 1.2 (A) Torque produced by the weight will restore the body to its original position. (B) Torque produced by the weight will topple the body. For simplicity, frictional forces that prevent the contact point from sliding are not shown. The frictional force is horizontal to the left in Figure (A) and to the right in Figure (B). (*See Chapter 2 for discussion of frictional forces.*)

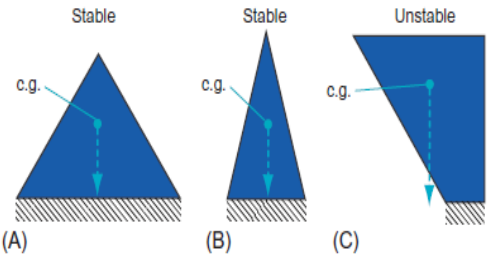


FIGURE 1.1 Stability of bodies.

- **2. Equilibrium in the Human Body**:
- The human body constantly adjusts its posture to keep the C.G. over the feet.

When this balance is lost (C.G. moves outside the base of support), the person

falls.

- **Compensation for Uneven Loads:**
  When carrying a weight in one arm, the torso bends away from the load,

and the other arm swings away from the body (Fig. 1.4). This shifts the C.G. back over the feet.

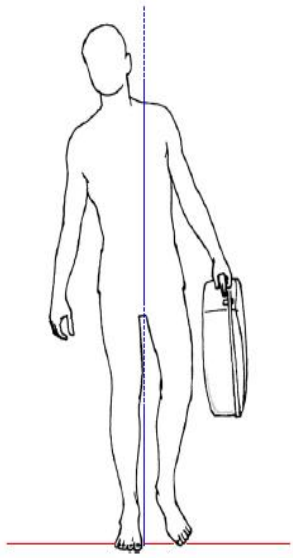


FIGURE 1.4 A person carrying a weight.

Effect of Amputation:
People who have lost an arm often bend continuously to compensate for the missing weight, which can lead to permanent spinal distortion. It is often recommended to wear an artificial arm, even if unused, to restore balanced weight distribution.

- Example: A force applied to the shoulder may topple a person standing at attention (Fig. 1.5).
  - **Data:**

Height of force application ($\boldsymbol{F_a}$) above ground:($\mathbf{1.5}$ m), Horizontal distance of C.G. from pivot point (A) (edge of foot): ($0.1\ m$), Person's mass (m): ($70\ Km$), Gravity (g): ($9.8\ \mathrm{m}/s^2$)

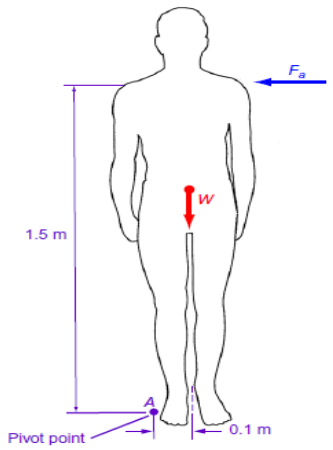


FIGURE 1.5 A force applied to an erect person.

- **Calculation:**

1. Person's weight (W):

$$\boldsymbol{w = m * g = 70\ Kg * 9.8 \frac{m}{s^2} = 686\ N}$$

2. Toppling torque (from external force F_a):

This torque acts counterclockwise about pivot A.

$$T_a = F_a = 1.5\ m$$

- **3. Stability Under an External Force:**

3. Restoring torque (from person's weight W):

This torque acts clockwise about pivot A.

$$T_\omega = W * 0.1\ m = 686\ N * 0.1 = 68.6\ N$$

4. Condition for impending toppling: Occurs when torques are equal:

$$\boldsymbol{T_a = T_\omega}$$

$$F_a * 1.5 = 68.6$$

$$F_a = \frac{68.6}{1.5} = 45.7\ N\ about\ 10.3\ Ib$$

- Conclusion: A relatively small sideways force (45.7 N) is theoretically sufficient to topple

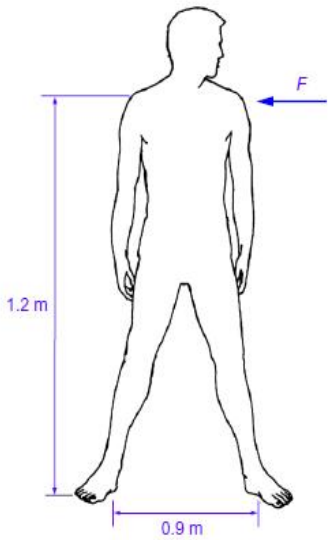
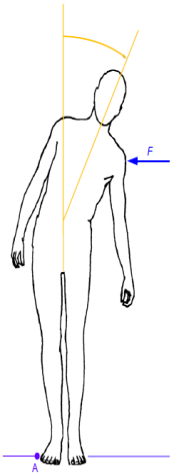


FIGURE 1.7 Increased stability resulting from spreading the legs.

a rigidly standing person.

- How does a person increase stability in practice?
  By bending the torso opposite to the applied force(Fig. 1.6). This shifts the C.G. away from pivot point (A), increasing the restoring torque due to body weight, thus raising



- FIGURE 1.6 Compensating for a side-pushing force.

  the force required for toppling.

- Widening the base of support: Spreading the legs (as in Fig. 1.7) increases stability against toppling forces.

- **General Conclusion:**

  1. Stable equilibrium requires the center of gravity to be above the base of support.
  2. The stability of the human body can be adjusted by:
     a. Modifying body posture (such as leaning or bending the torso).

b. Widening the base of support (such as by spreading the legs).

3. Relatively small forces can topple the body if their lever arm is long, but the body can withstand larger forces using compensatory mechanisms.

4. Understanding these principles is essential in fields such as rehabilitation, prosthetic design, and improving athletic performance.